

GENERAL EDUCATION: NATURAL SCIENCE

LET Competencies:

1. Utilize scientific knowledge in the preservation of human life and conservation of Earth's resources.
2. Relate scientific theories to actual and natural happenings that may induce calamities.
3. Utilize biological knowledge and theories in the improvement and preservation of the quality of life.
4. Utilize biological knowledge and theories in the improvement and preservation of the quality of life.
5. Apply scientific knowledge and principles for the improvement of human resources and socio-economic development.
6. Identify problems and suggest solutions related to personal and community.



CONTENT UPDATE

INTRODUCTION:

Scientific knowledge and skills are essential for economic and cultural development. Technology is very important to convert this scientific knowledge into outputs and concrete products. Science and technology has changed societies in pursuit of more comfortable lifestyle. The nanotechnology in telecommunications and innovations in transportations have drawn the world smaller and smaller. With the advent of information and communication technology, information can be relayed to other parts of the world in just a matter of seconds; the recent breakthroughs in genetic mapping have made it possible to identify the useful and bad genes that will result to the breeding of a better species. Transformations have not been entirely beneficial. Several technological innovations provide humans with a decent standard of living, but we see a lot of people living in poverty. There is also sufficient technology to produce enough food for all yet many are hungry and suffer from malnutrition. Technology has enabled humans to do things which were thought to be impossible before like exploring the universe, stem cloning, in-vitro fertilization, improving the means of communication yet they cannot solve the basic problems of food, health, and shelter. Science can be a vehicle to inculcate values on the proper use of technology that bring both advantages and disadvantages.

SCIENCE AND TECHNOLOGY:

SCIENCE is the pursuit for the profound knowledge. Scientists investigate the world around us. They observe how things work and develop ideas about ways to make them work better. Sometimes they try to test an idea to explain how something works. Scientists perform experiments to find out how things work. The knowledge that they discover is useful for many things. It can help to build new machines. It can help scientists to develop new medicines or cure a disease. Technology is the

SCIENCE is a systematized body of knowledge based on nature and the facts of life. The contents or facts of this knowledge are gathered through keen observation and relentless experimentations. The ideas generated are also products of further investigations. Science involves the human attempt to generate testable concepts, generalizations and theories which explain natural phenomena and have predictable capabilities.

TECHNOLOGY is the use of scientific knowledge to create new things. It is the application of scientific theories, laws, concepts, and principles. It is described as the means by which humans control or modify their environment which can be traced to Paleolithic cultures when humans shaped tools out of stone. Technology provides the eyes and ears of science.

SCIENTISTS share their basic beliefs and attitudes about the nature of science and technology.

WHAT ARE THE BASIC BRANCHES OF SCIENCE AND TECHNOLOGY?

A scientifically literate person has working knowledge of the basic concepts and principles of the natural sciences. Scientific knowledge is holistic, i.e., it is gathered by various branches of science contributing to an over-all conceptual scheme which is internally consistent.

- ❖ **Physical Science** – deals with the study of matter. This includes:
 - **Chemistry** – explore what substances are made of and how they change and combine.
 - **Physics** – explore the study of relationship between matter and energy.
- ❖ **Earth Science** – deals with the study of the Earth and its rocks, oceans, volcanoes, earthquakes, atmosphere, and other features. This includes astronomy, that explore nature beyond the Earth. This study objects such as stars, planets and moons.
- ❖ **Life Science (Biology)** - deals with living things and their parts and actions. Smaller branches of life science include:
 - **Zoology** – the study of animals
 - **Botany** – the study of plants

Science includes processes and products.

- ❖ **Processes**
 - manifests skills in solving problems, making decisions and furthering his/her own understanding of the universe
 - engages in activities that encourage higher levels of thinking and uses inquiry-oriented instruction
 - develops skills associated with the products of science and technology
- ❖ **Products** include the facts, concepts, theories, laws, and applications that occur as a result of doing science in a scientific content.

Scientific Facts: Below are some notions considered to be facts:

1. The earth rotates on its axis once every 24 hours.
2. The leaves are green due to its chlorophyll in the cells.
3. Water molecules are made of hydrogen and oxygen atoms.
4. A freely falling object accelerates toward the earth at the rate of 9.8 m/s^2

5. The temperature at which water freezes is 0° Celsius and boils at 100°C.

Scientific Concepts: These are ideas that combine several facts or observations. A concept is an "observed regularity in events or objects" (Novak, 1984).

1. Green plants need light in order to grow. - relates the two observations of light to the amount of growth of green plants
2. The human body uses food for energy and growth.
3. It takes more force to slide a book on sandpaper than on smooth paper.
4. Running water cuts gullies in soft rock.
5. Heavy marble roll farther than light marbles when both are given the same push.

Scientific Theories: Theories are developed to explain concepts or other observed regularities. These are proven hypothesis (tentative answers to questions).

1. The Cellular Theory of Life that states, all living things are made of cells.
2. The Kinetic Molecular Theory which states that molecules are always in motion, and the rate of motion varies with temperature.
3. Theory of Evolution which says the species adapt to their environments, and those that are most fit survive.
4. Theory of Plate Tectonic in which it states that the outer shell of the earth consists of several moving plates on which the oceans and continents lie.
5. The Dalton's atomic Theory which states that matter is made up of indivisible particles called atoms.

Scientific Laws are statements generally accepted to be true, universal, and absolute. They can be expressed in terms of a single mathematical equation. They are accepted always to be true.

1. Newton's Laws of Motion: Inertia, Acceleration and Interaction
2. The Gas Laws which relate pressure, volume and temperature of gases.
3. The Law of Universal Gravitation: All objects attract all other objects with a force that depends on their masses and the distance between them.
4. The Law of Conservation of Energy: Energy can neither be created nor destroyed; they can only be transformed from one form to another
5. The Law of Conservation of Matter and Energy: The sum of matter and energy in the universe is constant.

WHAT ARE THE PROCESSES OF SCIENCE AND TECHNOLOGY?

A scientifically literate person practices the basic and scientific skills called scientist. A practicing scientist uses the scientific method to search for cause and effect relationships in nature.

Steps in the Scientific Method

- **Identifying the problem** - Asking question based on the inference made to identify the problem related to the observation; an investigation usually begins with an observation of a phenomenon.

Scientists use their own senses, scientific gadgets or instruments that enhance those senses, to answer and prove something through keen observations.

- **Inferring** is explaining something based on observations.
 - **Measuring** allows us to be more accurate about what we observe.
 - Putting things into groups because they are alike, it is **classifying**.
- **Formulating hypothesis:** A hypothesis is a proposed explanation for a phenomenon. Normally, it is a general statement about the relationship between the manipulated variable and a responding variable in order to explain the questions ask.
 - **Identifying variable** - A variable is a factor that affects other factors in an experiment. In a scientific investigation, one has to identify all related variable to test the consistency in the experiment and to relate the manipulated variable to the responding variable.

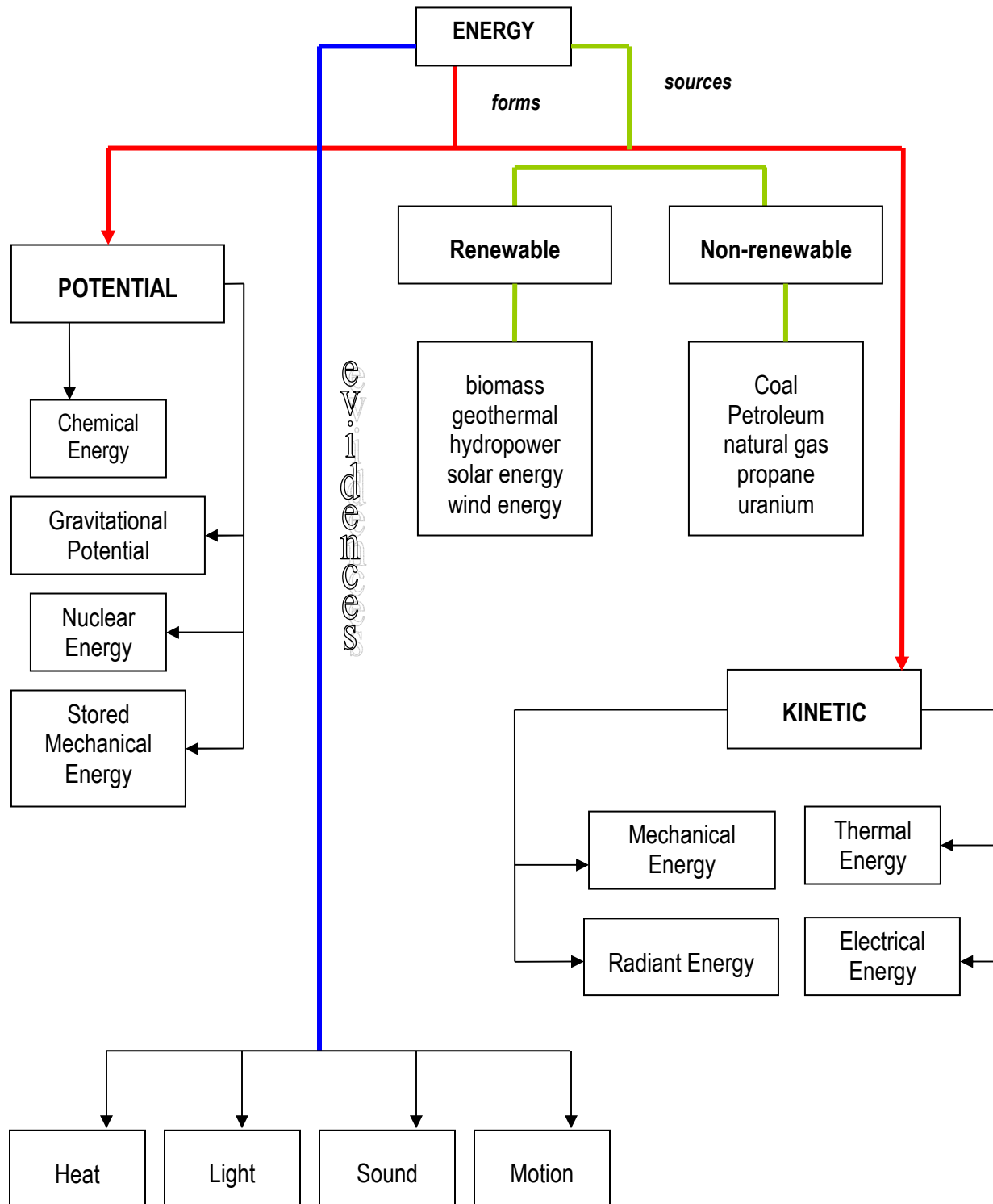
Three types of variable

- i. Manipulated variable — the factor that is purposely changed in an experiment
 - ii. Responding variable — the factor that changes with the manipulated variable
 - iii. Fixed variables — the factors that are kept constant throughout an experiment. This is to ensure that other factors do not affect the results of the experiment.
- **Testing the hypothesis** – the purpose of experimentation is to test the formulated hypothesis called scientific guess
 - **Designing and Experimenting**
When you test out your ideas why something happens you are **experimenting**. To carry out the experiment, one must prepare the materials, apparatus, experiment set ups and the procedures to take. It is important for the experiment to be a fair test by making sure that only one factor should be changed at a time while keeping all other conditions the same. Repeat the experiments several times to make sure that the first results weren't just an accident.
 - **Analyzing and Interpreting Data Gathered**
Studying results after doing an experiment and looking for relationships between the things being tested is **analyzing** and **interpreting** means making sense of science information presented in diagrams, flowcharts, tables, graphs, maps or words.
 - **Making Conclusion**
Draw conclusions based on the observations and results. State whether the hypothesis is true or false. **Applying** is using what you know or have found out in a new situation.

CONCEPTS IN PHYSICAL SCIENCE

A. ENERGY

- Energy is defined as the ability to do work. Energy makes changes and does things for us.
- Classification of energy



Energy is classified into **two main forms**:

A. **Kinetic energy** is defined as the energy of a moving object.

Example: Speeding cars, a waterfall, a rock falling from a cliff or the ticking of a clock are examples of objects that have kinetic energy.

Forms of Kinetic Energy:

- **Electrical energy** is the movement of electrical charges. All matter is made of tiny particles called atoms. These are made up of even smaller particles: electrons, protons, and neutrons. When a force is applied, some of the electrons move. Electrical charges moving through a wire is called electricity. Lightning is another example of electrical energy.
- **Mechanical energy** is the movement of objects and substances from one place to another when a force is applied according to Newton's Laws of Motion.
- **Thermal energy** is the internal energy in substances—the vibration and movement of the atoms and molecules within substances. This pertains to the kinetic energy of the molecules within an object. It is technically incorrect to refer to heat as thermal energy. Objects possess thermal energy; heat is the transfer of energy from one object to another. It is an evidence of energy.
- **Radiant energy** is electromagnetic energy that travels in transverse waves. Radiant energy includes visible light, x-rays, gamma rays and radio waves. Light is one type of radiant energy and is another evidence of energy.

B. **Potential energy** is defined as the energy in matter due to its position or the arrangement of its parts.

Forms of potential energy:

- **Chemical Energy** is energy stored in the bonds of atoms and molecules. It is the energy that holds these particles together.

Examples:

- The energy stored in gasoline is released by burning.
- A motorboat uses this released energy to turn its propeller.
- Energy in food is used by our bodies to move.
- A lighted firecracker explodes with a loud sound when chemical energy is released.

- **Gravitational Energy** is the energy of position or place.

Examples:

- A rock resting at the top of a hill contains gravitational potential energy.
- Water in a reservoir behind a dam, is an example of gravitational potential energy.

- When something is lifted or suspended in air, work is done on the object against the pull of gravity. The work done is converted gravitational potential energy.
- **Stored Mechanical Energy** also known as elastic potential energy is the energy in stretched rubber bands, in compressed springs, trampolines, and even our skin. They have the potential to do work or resist being stretched out of shape.
- **Nuclear Energy** is energy stored in the nucleus of an atom and can be released when the nuclei are combined or split apart. Nuclear power plants split the nuclei of uranium atoms in a process called **fission**. The sun combines the nuclei of hydrogen atoms in a process called **fusion**.

Sources of Energy:

- ❖ **Renewable energy** is the energy that can be replaced over a very short period of time. This includes biomass, geothermal energy, hydropower, solar energy, and wind energy. They can be replenished in a short time.
- ❖ **Nonrenewable energy** supplies are limited and can be replaced in a long period of time. Petroleum, for example, was formed millions of years ago from the remains of ancient sea plants and animals.

Evidences of Energy:

- ❖ **Light** is an electromagnetic radiation and has no mass, hence, it has neither kinetic nor potential energy.
- ❖ **Sound** is produced when a force causes an object or substance to vibrate—the energy is transferred through the substance in a wave.
 - The human ear can hear only those sounds caused by objects vibrating between about 20 vibs/sec and about 16,000 vibs/sec.
 - Ultrasonic waves are sound waves that vibrate very fast with frequencies too high to be detected by the human ear. They greater penetrating power than audible sound waves and can be used to see *in utero* fetuses and the functioning of human hearts.
- ❖ **Heat** consists of moving molecules in air or in an object, and mechanical energy is the combination of kinetic and potential energy of a moving object. A pendulum has mechanical energy; it continually converts kinetic energy into gravitational potential energy and back into kinetic energy as it swings back and forth.

Transformation of energy – interconversion of energy from one form to another

- The law of conservation of energy says that energy is neither created nor destroyed. When we use energy, it doesn't disappear. We change it from one form of energy into another.
- Converting one form of energy into another form always involves a loss of usable energy. The total amount of energy available for transformation is almost always decreasing.

- When energy is transformed into heat energy that diffuses all over, further transformations is less likely to occur. Energy follows a one-way path. It enters the living system in the form electromagnetic waves - light and leaves as heat.

Examples of Energy Transformations:

- Stored energy in a flashlight's batteries becomes light energy when the flashlight is turned on.
- Food is stored energy. It is stored as a chemical with potential energy. When your body uses that stored energy to do work, it becomes kinetic energy.
- If you overeat, the energy in food is not "burned" but is stored as potential energy in fat cells.
- When you talk on the phone, your voice is transformed into electrical energy, which passes over wires (or is transmitted through the air). The phone on the other end changes the electrical energy into sound energy through the speaker.
- A car uses stored chemical energy in gasoline to move. The engine changes the chemical energy into heat and kinetic energy to power the car.
- A toaster changes electrical energy into heat and light energy. (If you look into the toaster, you'll see the glowing wires.)
- A television changes electrical energy into light and sound energy.

B. Structure Of Matter

Matter is anything that occupies space and has mass. They differ greatly in shape, density, flexibility, texture, toughness, and color; in their ability to give off, absorb, bend, or reflect light; in what form they take at different temperatures; in their responses to each other; and in hundreds of other ways.

Forms of Matter

❖ Solid

- has definite shape and size.
- molecules are packed very close to each other
- retains their own shape.

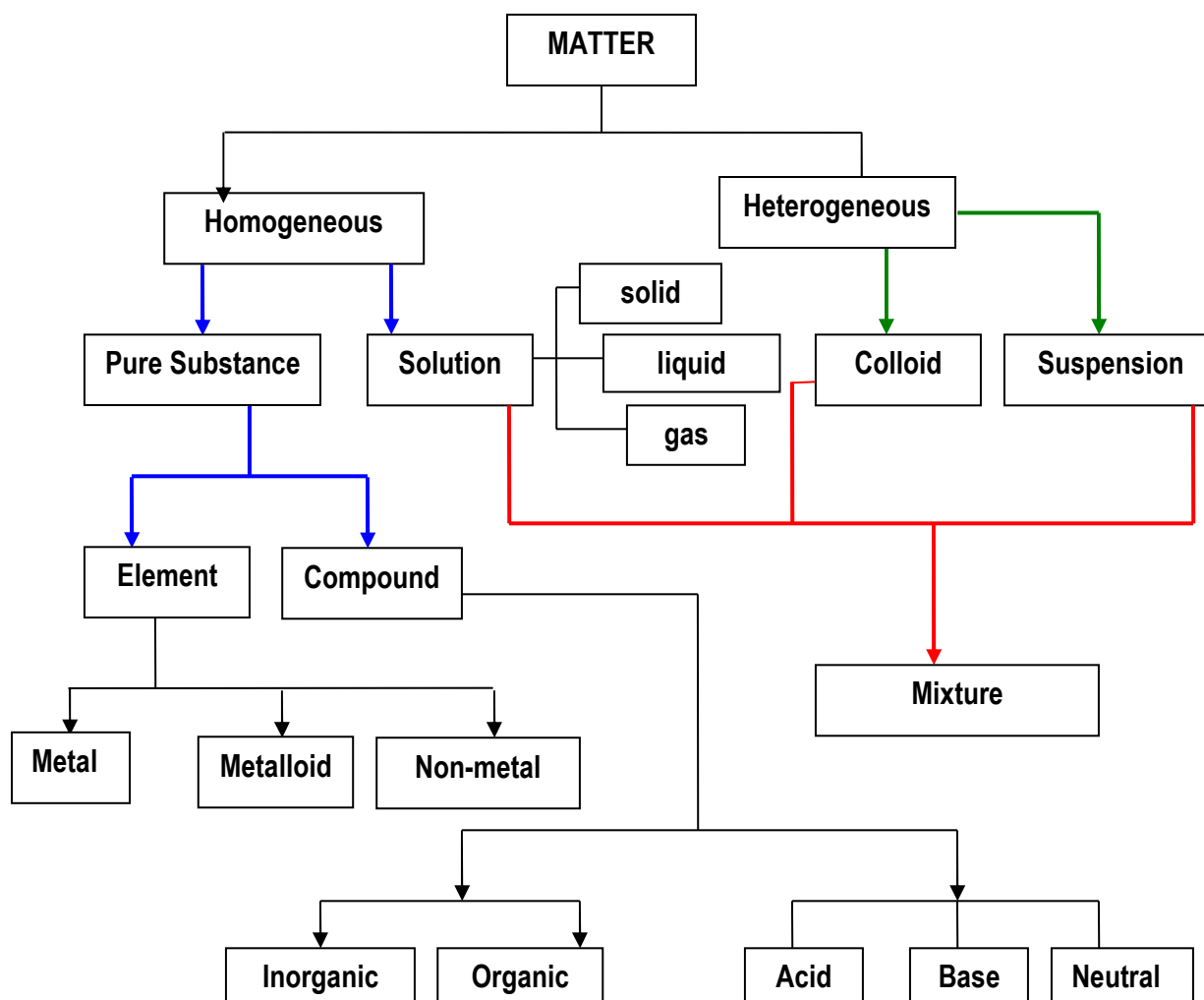
❖ Liquid

- does not have definite shape but have definite size.
- acquires the shape of the container.
- molecules of liquids are farther apart than in solids so they can move more freely.
- are fluid because they can flow.

❖ Gas

- can be squeezed into containers like balloons,
- if released, spread into a wider space.
- the molecules are very far apart.
- does not have definite shapes and sizes because particles are free to spread out at all directions

Classification of matter according to composition:



- ❖ **Homogeneous system** is characterized by a one-phase system whose characteristics are uniform throughout. These are observed either as a pure substance or in solutions. An example of a homogeneous substance would be pure water, which contains only the compound H_2O or pure table salt that contains only the compound NaCl . Two or more gases will always mix to form just one phase.
 - **Elements** are the simplest form of matter since they cannot be decomposed any further by any other ways. Elements are further classified into metal, non-metal and metalloid. Some characteristics of **metal**: luster, malleability and ductility, high tensile strength, good conductors of heat and electricity. **Nonmetals** are characterized as are poor conductors (thus, making them good insulators) of heat and electricity, neither malleable nor ductile, brittle. (if in the solid phase)
 - **Compounds** are pure substances that are composed of two or more elements, which are chemically combined in a definite ratio by mass. There are two ways of grouping compounds – as organic or inorganic; and as acid, base or neutral. Basically, carbon-containing compounds are organic. Sugar, acetone, acetic acid, methane, ethylene,

alcohol are organic compounds. However, carbides, carbonates, bicarbonates, cyanides, carbon dioxide and carbon monoxide are exemptions.

- **Acids** are usually distinguished from **bases** through the use of indicators. **Indicators** are either weak organic acids or bases that change color over a range of pH values. Litmus is a common indicator. Acid turns the color of blue litmus to red while base turns red to blue. Bases turn red litmus paper to blue. Acids when dissolved in water yield hydrogen ions (H^+) while bases yield hydroxide ions (OH^-). Unlike acids and bases, neutral compounds do not change the color of indicators.
 - **Solution** is the only homogeneous **mixture**, made up of two or more pure substances that are physically combined. Alloys are solid solutions which are combinations of metals. Salt and water forms a liquid solution while air is a mixture of gases. Solutions contain a *solvent* (the material that does the dissolving of another material) and a *solute* (the material that is dissolved). Water is called the “universal solvent” because it dissolves many substances. The warmer the solvent, the more solute that can be dissolved.
- ❖ **Heterogeneous system** is a **mixture** of two or more substances combined in no definite proportions.
- **Colloids** and **suspensions** are heterogeneous. Of the three types of mixtures, only **colloids exhibit Tyndall effect**. Tyndall effect is the scattering of light.
 - **Suspensions** are mixtures of a solid, liquid and/or gas materials. Examples of suspensions are soil particles in water, halo-halo, oil in water.

Ways of Separating Mixtures:

- ❖ The components of mixtures can be separated by **ordinary physical processes**. The choice of separation technique is dependent on the type of mixture and the characteristics of its components.
- **Filtration** is a separation technique that is being used in separating insoluble material from the liquid by allowing the mixture to pass through a filter paper.
 - **Decantation** is a process for the separation of mixtures, carefully pouring a solution from a container, leaving the precipitate (sediments) in the bottom of the container. Usually small amount of solution must be left in the container, and care must be taken to prevent a small amount of precipitate from flowing with the solution out of the container.
 - **Centrifugation** is a separation process which uses the action of centrifugal force to promote accelerated settling of particles in a solid-liquid mixture. Two distinct major phases are formed in the vessel during centrifugation the **sediment** usually does not have a uniform structure and the supernatant liquid
 - **Sedimentation** is the tendency for particles in suspension to settle out of the fluid in the mixture

- Two immiscible liquids can be separated by **using a separatory funnel** and a mixture that is composed of a metal and a non-metal can be easily separated by **using a magnet**.
- A mixture that is made up of two solids can be separated either by **using a sieve or by dissolution**. On the other hand, soluble solids from a liquid can be separated through **evaporation or distillation**.

Properties of Matter

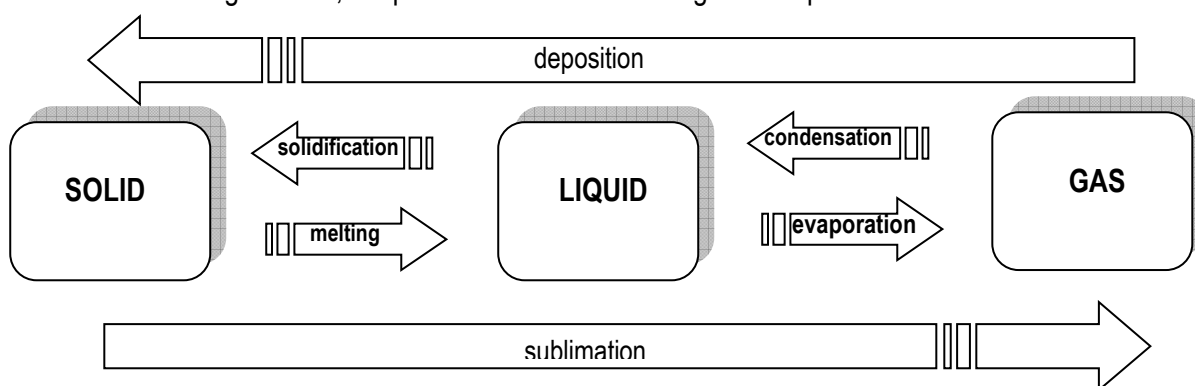
- Some properties, such as mass and volume, depend on the **quantity** of matter is called **extensive** while those that do not depend and distinguish each kind of matter from one another like density is termed as **intensive**.

Changes in Matter

- **Physical Change** – a change in matter that can be observed or measured without changing the identity includes:

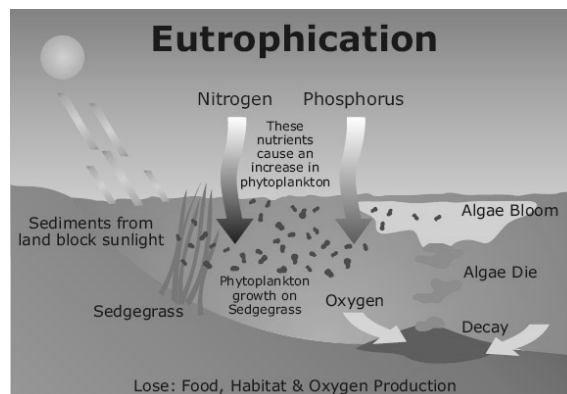
change in the state of matter

change in size, shape or state without a change in composition

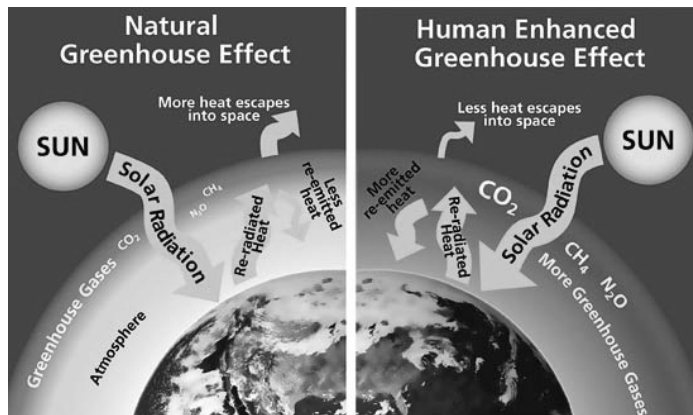


- **Chemical Change** – a change in matter resulting in the formation of new identity or one or more substances
- Mixtures of substances in the waters, land, and air may result to some major environmental problems:
 - **Water Pollution** - contamination of water by foreign matter such as microorganisms, chemicals, industrial or other wastes or sewage.

- **Eutrophication** - a natural or artificial process on which a body of water contains high concentrations of chemical elements required for life making the producers bloom directly nourishing most bacteria than aquatic organisms like fishes. This results to red tides.



- **Green House Effect:** a process in which sunlight is not capable of escaping the earth's atmosphere due to suspended particles in air resulting to global warming



Force and Motion

Force is a push or pull upon an object resulting from the object's *interaction* with another object. They can either be contact or non-contact force.

Contact Forces	Action-at-a-Distance Forces
Friction	Gravitational Force
Tension	Electrical Force
Normal Force	Magnetic Force
Air Resistance	
Applied Force	
Spring Force	

- **Contact forces** result when two bodies directly touch each other as in walking, running, playing, writing and pushing objects where friction is observed.

Types of Contact Forces

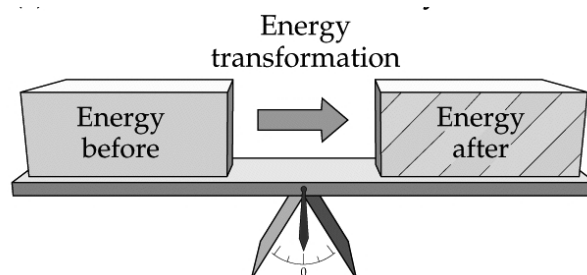
- **Frictional Force** is a force that opposes the natural motion of an object. It is often described as the resistance an object meets when its surface rubs against another surface. It acts in a direction opposite the natural motion of the moving object.
 - There are 3 types of frictional forces depending on the type of motion that occurs between two surfaces. **Static friction** - if there is no relative motion between two surfaces; **kinetic friction** - the type of friction that opposes sliding motion, is weaker than static friction and the friction force that exists in rolling motion is called **rolling friction**. This is the weakest frictional force that opposes motion.
- **Tension** - The tension is the force which is transmitted through a string, rope, or wire when it is pulled tight by forces acting from each end. The tensional force is directed along the wire and pulls equally on the objects on either end of the wire.

- **Normal force** - The normal force is the support force exerted upon an object which is in contact with another stable object. For example, if a book is resting upon a surface, then the surface is exerting an upward force upon the book in order to support the weight of the book. On occasions, a normal force is exerted horizontally between two objects which are in contact with each other.
 - **Air resistance** - The air resistance is a special type of frictional force which acts upon objects as they travel through the air. Like all frictional forces, the force of air resistance always opposes the motion of the object. This force will frequently be neglected due to its negligible magnitude. It is most noticeable for objects which travel at high speeds (e.g., a skydiver or a downhill skier) or for objects with large surface areas.
 - An **applied force** is a force which is applied to an object by a person or another object. If a person is pushing a desk across the room, then there is an applied force acting upon the object. The applied force is the force exerted on the desk by the person.
 - **Spring force** - The spring force is the force exerted by a compressed or stretched spring upon any object which is attached to it. An object which compresses or stretches a spring is always acted upon by a force which restores the object to its rest or equilibrium position.
- **Non-contact forces** are forces that occur when the fields around objects interact with another field located around another body. The bodies themselves are not directly touching each other, but only their fields interact with one another.

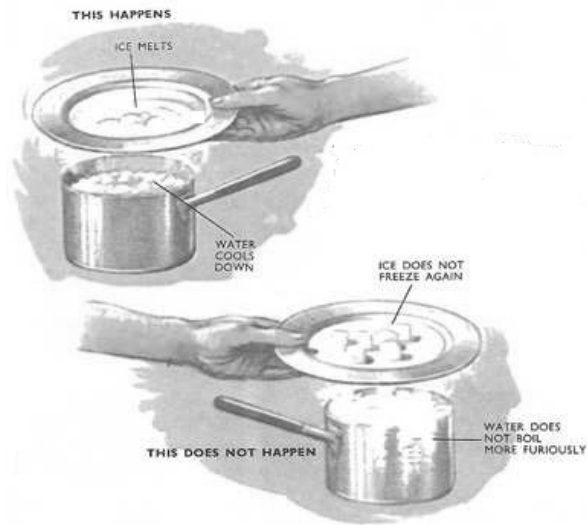
Types of Non-contact Forces

- **Gravitational force** is observed when earth is attracted to the sun, earth also pulls the sun. An object falling freely in the air is acted upon by a force of gravity that cause it to accelerate. ($g = 9.8 \text{ m/s}^2$)
 - Gravity is the force behind rainfall, the power of rivers, the pulse of tides; it pulls the matter of planets and stars toward their centers to form spheres, holds planets in orbit, and gathers cosmic dust together to start the process of star formation.
 - Gravitational forces involve a gravitational field that affects space around any mass. The strength of the field around an object is proportional to its mass and diminishes with distance from its center. ($F = G m_1 m_2 / r^2$) where $G = 6.67 \times 10^{-11} \text{ Nm/kg}^2$
- **Electrostatic force** is the interaction between the electric field of one charged body another charged body
 - On an atomic scale, electric forces between oppositely charged protons and electrons hold atoms and molecules together
 - An electrically insulating material such as glass or rubber does not ordinarily allow any passage of charges through it. On the other hand, an electrically conducting material such as copper will offer very little resistance to the motion of charges, so electric forces acting on it readily produce a current of charges. (Most electrical wires are a combination of extremes: a very good conductor covered by a very good insulator.)

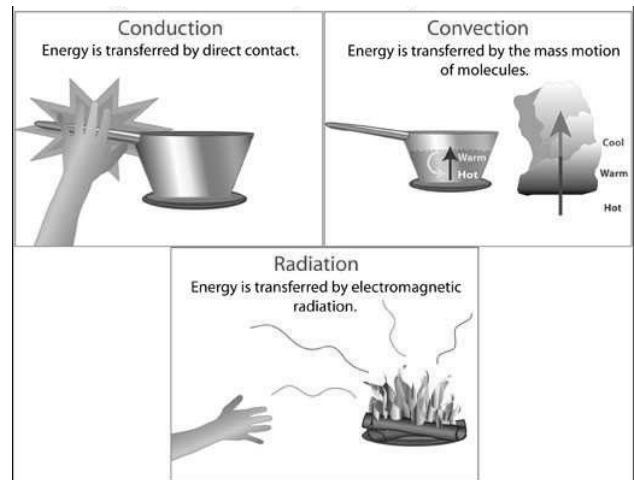
- **Magnetic force** is the interaction between magnetic fields pulling or pushing each other – like poles attract each other, unlike poles, repel each other.
 - Magnetic forces are very closely related to electric forces—the two can be thought of as different aspects of a single electromagnetic force. Both are thought of as acting by means of fields: an electric charge has an electric field in the space around it that affects other charges, and a magnet has a magnetic field around it that affects other magnets.
 - Moving electric charges produce magnetic fields and are affected by magnetic fields. This influence is the basis of many natural phenomena. For example, electric currents circulating in the earth's core give the earth an extensive magnetic field, which we detect from the orientation of our compass needles.
- The interplay of electric and magnetic forces is also the basis of many technological design, such as electric motors (in which currents produce motion), generators (in which motion produces currents), and television tubes (in which a beam of moving electric charges is bent back and forth by a periodically changing magnetic field). More generally, a changing electric field induces a magnetic field, and vice versa.
- When an unbalanced force does act on an object, the object's motion changes. It may change its speed or its direction of motion or both. The greater the amount of unbalanced force, the more rapidly a given object's speed or direction of motion changes; the more massive an object is, the less rapidly its speed or direction changes in response to any given force.
- **Sir Isaac Newton** developed three **laws of motion** that explains force and motion.
 - **1st Law of Motion** - an object at rest remains at rest and an object in motion remains in motion in a straight line and at constant speed unless acted upon by an external unbalanced force. (Inertia) ex: This is the reason for seat belts and air bags as safety devices in cars.
 - **2nd Law** – force is directly proportional to mass and acceleration ($F = ma$)
 - **3rd Law** states that for every force there is an equal and opposite force. This is the familiar "action-reaction" system and it governs the movement of rockets and jet airplanes
- **Work and Energy** – when an object does work on another object, energy is transferred
 - Work output is always less than the work input because of friction. When work is done, heat is always dissipated.
 - Disregarding friction, work output is equal to the work input
 - Heat is the result of heat transfer. It is the measure of the total kinetic energy of an object.
 - Temperature is the measure of the average kinetic energy of an object
 - Thermodynamics – the study of the effects of work, heat and energy on a system.
 - Laws of Thermodynamics
 - First Law – conservation of energy



- Second Law – heat always flows from hot to cold and not from cold to hot.



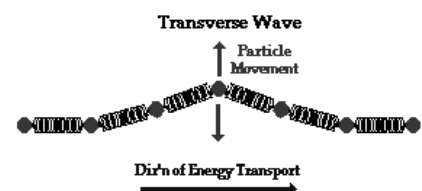
- Methods of heat transfer
 - Conduction – through solids
 - Convection – through fluids
 - Radiation – through space or vacuum



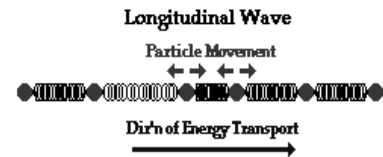
- Some complicated motions can be described most conveniently not in terms of forces directly but in terms of the pattern of motion, such as vibrations and waves. Examples are sound waves, light waves, earthquakes, or surface waves on water.
- Wave behavior can be described in terms of how fast the disturbance propagates (speed, v), and in terms of the distance between successive peaks of the disturbance (wavelength, λ). Wavelength can greatly influence how a wave interacts with matter—how well it is transmitted, absorbed, reflected, or diffracted.

- Types of Waves:

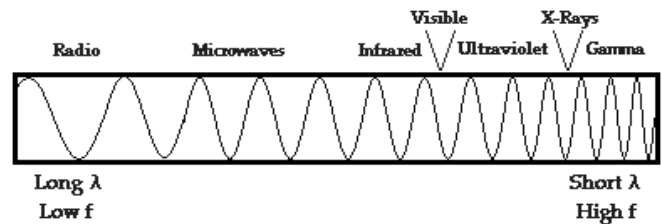
- **Mechanical waves** – a disturbance that travels through a medium. The particles in the medium undergo displacements of various kinds depending on the nature of the wave.



- A transverse wave is a wave in which particles of the medium move in a direction perpendicular to the direction which the wave moves.
- A longitudinal wave is a wave in which particles of the medium move in a direction parallel to the direction which the wave moves.



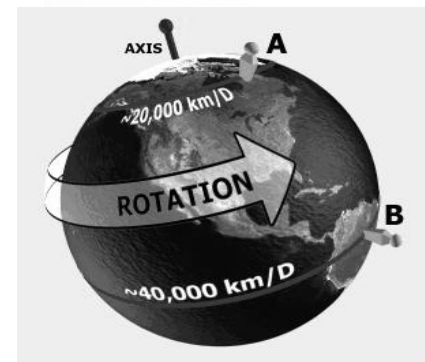
- **Electromagnetic waves** – waves which do NOT require a material medium for their transmission. This includes the regions from longest wavelength—radio waves, microwaves, radiant heat or infrared radiation, visible light (red, orange, yellow, green, blue, indigo, violet), ultraviolet radiation, x rays, and gamma rays – shortest wavelength.



- Atmospheric gas, ozone, absorbs some of the ultraviolet radiation in sunlight—the wavelengths that produce burning, tanning, and cancer in the skin of human beings.
- Ultrasonic waves are sound waves that vibrate very fast. Their frequencies are too high to be detected by the human ear. Ultrasonic sound waves have greater penetrating power than audible sound waves and can be used to see *in utero* fetuses and the functioning of human hearts.
- Doppler effects are evident in the apparent change in pitch of an automobile horn as it passes the observer. The change in the wavelength of light from stars and galaxies also illustrates Doppler. Because the light emitted from most of them shifts toward longer wavelengths (that is, toward the red end of the spectrum), astronomers conclude that galaxies are all moving away from one another—and hence that we are in a generally expanding universe. (Doppler shift)

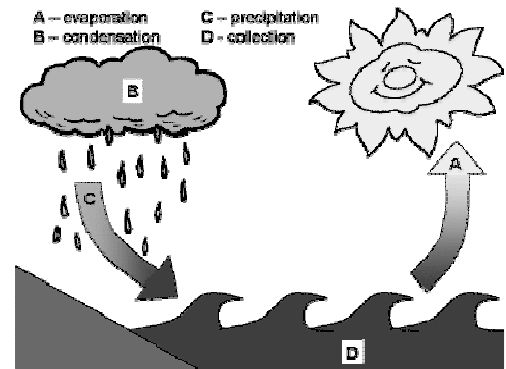
EARTH AND SPACE SCIENCE

- Earth is a fairly small planet. The distance of the earth from the sun ensures that energy reaches the planet at a rate sufficient to sustain life. The earth is mostly rock, with three-fourths of its surface covered by a relatively thin layer of water and the entire planet enveloped by a thin blanket of air. Bulges in the water layer are raised on both sides of the planet by the gravitational pull of the moon and sun, producing high tides about twice a day along ocean shores. Similar bulges are produced in the blanket of air as well.
- The earth is approximately 1.28×10^4 km in diameter at the equator. The diameter at the poles is approximately 70 km less than at the equator because of its rapid rotation on its axis (equatorial bulge).
- The motion of the earth and its position with regard to the sun and the moon has noticeable effects. The earth's one-year revolution around the sun, because of the tilt of the earth's axis, changes how



directly sunlight falls on one part or another of the earth.

- This difference in heating different parts of the earth's surface produces seasonal variations in climate. The rotation of the planet on its axis every 24 hours produces the planet's night-and-day cycle. The combination of the earth's motion and the moon's own orbit around the earth, once in about 27 1/3, results in the phases of the moon (new moon, 1st quarter, full moon, last quarter).
- Transfer of heat energy at the interfaces of the atmosphere with the land and oceans produces layers at different temperatures in both the air and the oceans. These layers rise or sink or mix, giving rise to winds and ocean currents that carry heat energy between warm and cool regions. The earth's rotation curves the flow of winds and ocean currents, which are further deflected by the shape of the land.
- The water cycle plays an important part in determining climatic patterns—evaporating from the surface, rising and cooling, condensing into clouds and then into snow or rain, and falling again to the surface, where it collects in rivers, lakes, and porous layers of rock. There are also large areas on the earth's surface covered by thick ice (such as Antarctica), which interacts with the atmosphere and oceans in affecting worldwide variations in climate.
- Fresh water is an essential resource for daily life and industrial processes, obtained from rivers and lakes and from water that moves below the earth's surface (groundwater). Many sources of fresh water cannot be used because they have been polluted. It can be very expensive to clean up polluted air and water, restore destroyed forests and fishing grounds, or restore or preserve eroded soils of poorly managed agricultural areas.
- Although the oceans and atmosphere are very large and have a great capacity to absorb and recycle materials naturally, they do have their limits. They have only a finite capacity to withstand change without generating major ecological alterations that may also have adverse effects on human activities. Wind, tides, and solar radiation can also be harnessed to provide sources of energy.
- Forces within the earth cause continual changes on its surface. The solid crust of the earth—including both the continents and ocean basins—consists of separate sections that overlie a hot, almost molten layer.

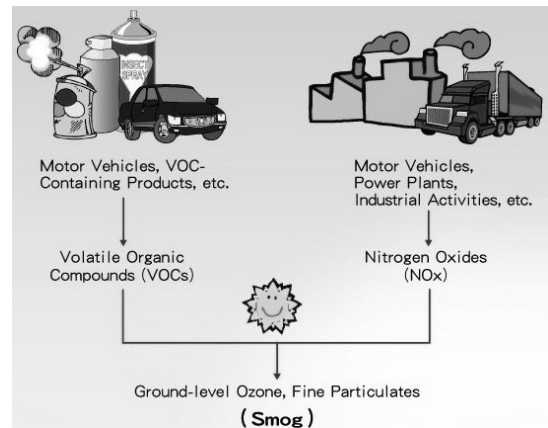


- The theory of plate tectonics is a unifying theory of the continental drift and seafloor spreading theories. The current position of continents on the globe is thought to be the result of the break-up and subsequent movement of the plates making up the super continent, Pangea, about 250 million years ago. Pangea is formed from the collision of continents that formed from the original crust of the earth and started moving around 2 1/2 billion years ago.
- The surface of the earth undergoes continuous cycles of building structures and tearing them down. Weathering is process of wearing structures down while erosion refers to



moving the weathered materials. Canyons, gullies, and valleys are the result of weathering and erosion by rivers and streams.

- Human activities have changed the earth's land surface, oceans, and atmosphere. For instance, reducing the amount of forest cover on the earth's surface has led to a dramatic increase in atmospheric carbon dioxide, which in turn may be leading to increased average temperature of the earth's atmosphere and surface.
- Smoke and other substances from human activity interact chemically with the atmosphere and produce undesirable effects such as smog, acid rain, and an increase in the damaging ultraviolet radiation that penetrates the atmosphere.
- Intensive farming has stripped land of vegetation and topsoil, creating virtual deserts in some parts of the world.



The Universe

- The earth has existed for only about a third of the history of the universe and is in comparison a mere speck in space.
- Our sun is a medium-sized star orbiting near the edge of the arm of an ordinary disk-shaped galaxy of stars, the Milky Way. Our galaxy contains many billion stars, and the universe contains many billion such galaxies.
- The entire contents of the known universe expanded explosively into existence from a single hot, dense, chaotic mass more than ten billion years ago. Stars coalesced out of clouds of the lightest elements (hydrogen and helium), heated up from the energy of falling together, and began releasing nuclear energy from the fusion of light elements into heavier ones in their extremely hot, dense cores.
- There are also a great many smaller bodies of rock and ice orbiting the sun. Some of those that the earth encounters in its yearly orbit around the sun glow and disintegrate from friction as they plunge into the atmosphere—and sometimes impact the ground.



- Other chunks of rock mixed with ice have such long and off-center orbits that they periodically come very close to the sun, where some of their surface material is boiled off by the sun's radiation and pushed into a long illuminated tail that we see as a comet.
- In studying the universe, a variety of tools are used. These tools include radio and x-ray telescopes that are sensitive to a broad spectrum of information coming to us from space. Computers can undertake increasingly complicated calculations of gravitational systems or nuclear reactions, finding patterns in data and deducing the implications of theories. Space probes send back detailed pictures and other data from distant planets in our own solar system; and huge "atom smashers" that simulate conditions in the early universe and probe the inner workings of atoms.

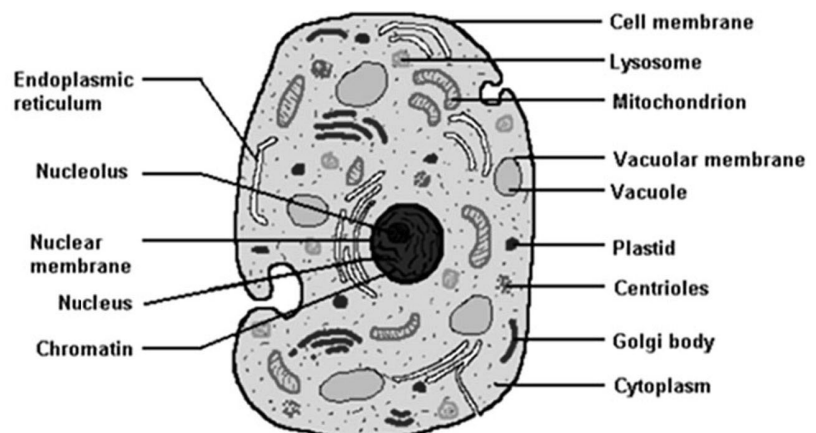


CONCEPTS IN BIOLOGICAL SCIENCE

Living organisms are made of the same components as all other matter, involve the same kind of transformations of energy, and move using the same basic kinds of forces.

The Cellular Theory of Life

- Living things are made of cells. Some organisms such as bacterium, an amoeba and a paramecium, are made of single cells (unicellular organisms); others are multicellular.
- Complex organisms are made of different kinds of cells specialized to carry out specific functions, all working together to accomplish the total needs of life (Ex: cells specialized for growing hair, secreting digestive juices, making bones and muscles, and carrying oxygen)
 - A group of similar cells performing similar functions form tissues (ex. are human tissues are muscle tissue, blood tissue and bone tissue).
 - A group of tissues working together forms an organ. Examples of human organs are stomach, the biceps muscle, and the heart.
 - Several organs working together form a system. All the systems make up the organism. Examples of human systems include the circulatory system, the digestive system, and the muscular system. Plant systems include the root system, leaf system and the stem system.
- The cell is the fundamental unit of life, made up of a protoplasm, enclosed by a plasma membrane, and containing organelles.
 - The plasma membrane provides form and structure for the cell, and permits inward passage of waste products.
 - The nucleus is the control center of the cell

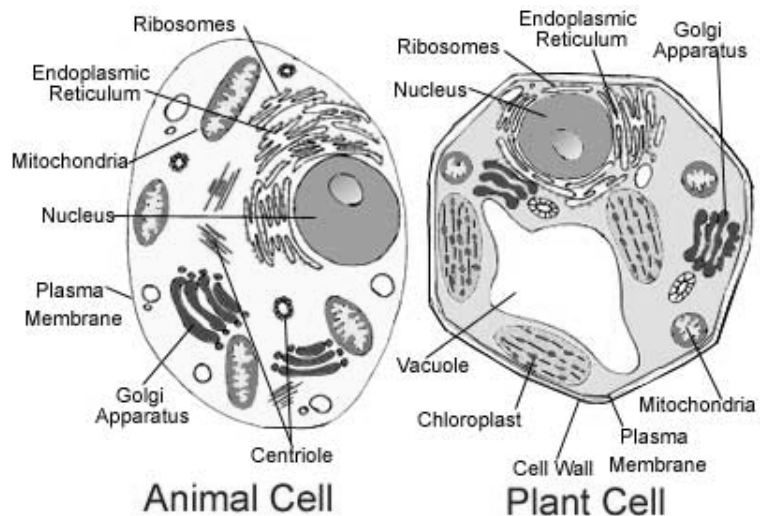


and contains the chromosomes, which are made of genes, which, in turn, are made of DNA. Most directions for cellular activity are given by the DNA in the nucleus.

- The cytoplasm in the cell includes all the material outside the nucleus.

- In plants cells, the plasma membrane is surrounded by a cell wall made of cellulose to provide rigidity for the cells because plants do not have a skeletal structure.

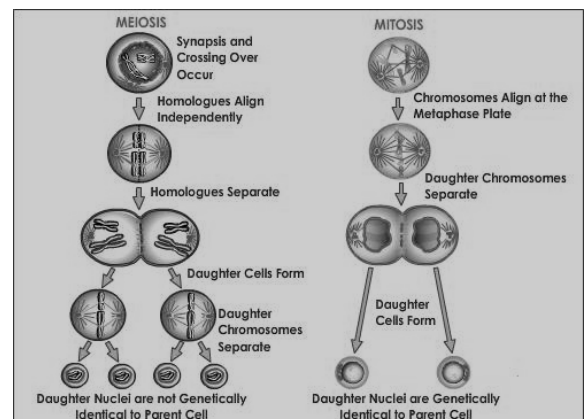
- Plant cells also contain plastids, tiny colored bodies that provide color and are used for storage or as chemical factories. Chloroplasts are plastids that contain chlorophyll, which is used in making food. When the green leaves die in autumn, the chlorophyll fades and other plastids can be seen, resulting in the brightly colored fall foliage.



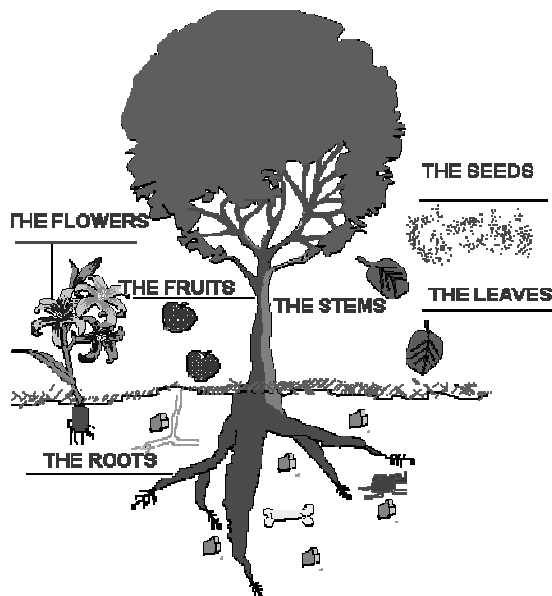
- Cells have basic living functions, some of which are: (1) to manufacture proteins and other materials to help build the cell; (2) to manufacture energy; and (3) to reproduce.
 - Green plants have the unique capability of manufacturing their own food in the presence of light from water, minerals, and carbon dioxide in a process called photosynthesis. The products of this reaction are glucose (energy), H_2O , and O_2 . (Photosynthesis)
 - Both oxygen and glucose are used in energy production. All cells, including animal and plant cells, use oxygen for manufacturing energy and release carbon dioxide as a waste product. Thus, oxygen is used by both plants and animal cells, and carbon dioxide is given off by both plants and animal cells.

- Cells reproduce either by mitosis (somatic cells) or meiosis (sex cells)

- Mitosis produce two new cells whose chromosomes are the same in number as the parent cell (diploid)
- Meiosis produce two new cells with half the number of chromosomes as the parent cell (haploid)



Structure and Function of Plants

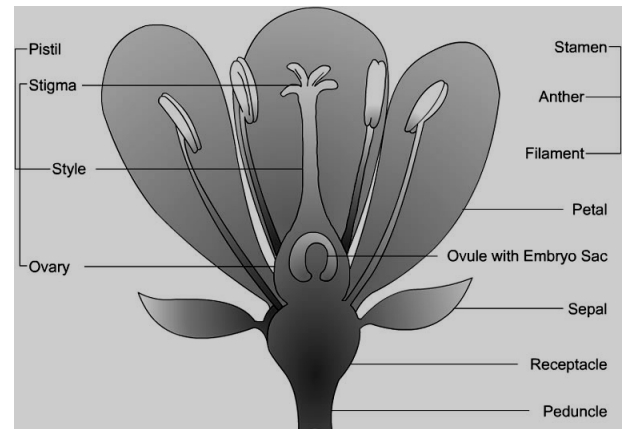


➤ **Plants** contain organs called roots, stems, leaves, and flowers.

- The roots provide anchorage for the plant and, through tiny root hairs, enable plants to absorb water and nutritional elements from the soil.
- The stems transport the nutrients to the leaves through cells that are specialized for transport.
- The leaves are the food-manufacturing sites of the plants.

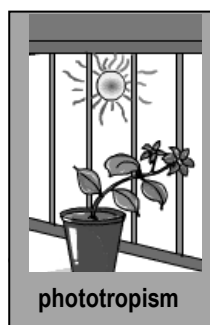
➤ **Flowers** are the sexual reproductive organs of the plants.

- The petals attract insects and birds to help disperse pollen for cross-fertilization.
- The male reproductive organ is the stamen, which manufactures the pollen.
- The female reproductive organ is the pistil, which has a sticky top called a stigma to capture the pollen; a style to transport the pollen nucleus to the ovary; and the ovary, which manufactures the ovules.
- When the pollen nuclei join the ovule nuclei, fertilized seeds result.



➤ Plants move in response to gravity, light, touch, water, temperature and some chemicals

- Roots pointing downward shows plants respond to gravity (**geotropism**).



- Plants also tend to grow toward areas with the most light, causing them to bend; houseplants bend toward the light and forest plants grow tall, reaching for the sun. (**phototropism**)



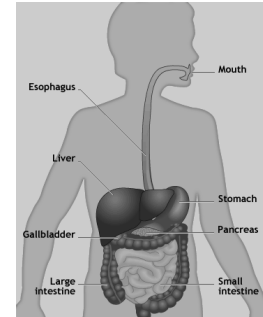
- Some plants, (mimosa) have leaves that quickly close when touched (**thigmotropism**).



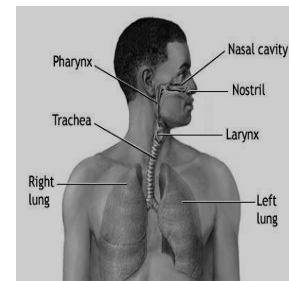
Structure and Function of Animals

- The basic functions of all animals include nutrition, respiration, excretion, circulation, response, movement, regulation, and reproduction.

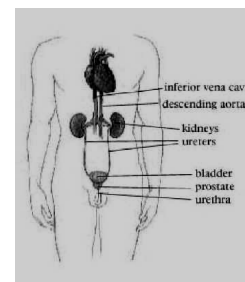
- The digestive system takes food in, processes it for use by cells, and eliminates indigestible materials.
 - The digestive system is made up of many important organs that work together to digest the food you eat. It also helps the body absorb all the nutrients that the food contains. Nutrients are substances that keep you healthy.



- The respiratory system is adapted for extracting the required oxygen either from the air or from water and removing carbon dioxide. It helps your body take oxygen from the air. The respiratory system is made up of your nose, mouth, trachea, lungs, and diaphragm

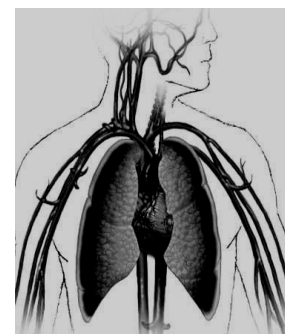


- Excretion is accomplished through kidneys and other excretory organs that extract liquid waste from body fluids and eliminate it from the organism. Made possible through the **KIDNEYS** and other excretory organs that extract liquid waste from the body fluids and eliminate from the organisms



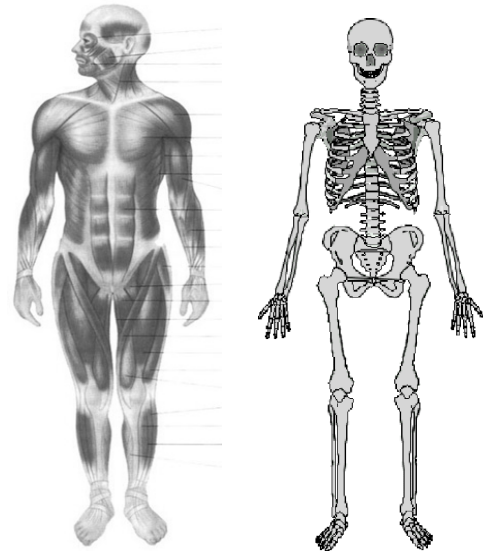
- A complex circulatory system ensures that nutrients, gases, and regulatory hormones are transported from the specialized systems to each individual cell and that wastes can be removed from each cell and delivered to the excretory and respiratory organs.

- The **CIRCULATORY SYSTEM** is made up of the **HEART** and blood vessels known as arteries and veins. The heart pumps blood throughout your body through the blood vessels. Blood delivers oxygen and nutrients to cells and carries away carbon dioxide and other waste materials.



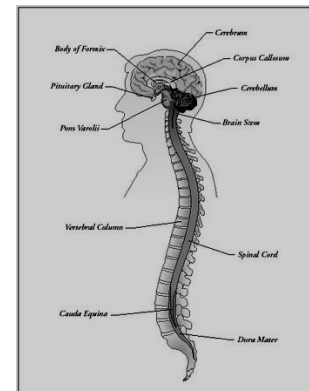
- Regulatory glands secrete hormones to keep all cells working harmoniously.

- The **muscular system** allows the organism to move. Human has 650 muscles in the body. Some muscles work by themselves to digest food or make your heart beat. Other muscles are used to do such things as kicking a ball or lifting a bag of groceries.
- A **skeletal system** provides rigidity in animals that do not otherwise have the required structure. Skeletons may be made of bone, cartilage, or water (such as hydroskeletons in certain worms). Some animals, such as crayfish and lobsters, have exoskeletons (external skeletons) that are made of chitin. New born babies have 295 bones while adult skeletal system contains **206** bones of various shapes and sizes. Bones protect your internal organs and support your body as you stand, sit, walk, and run.



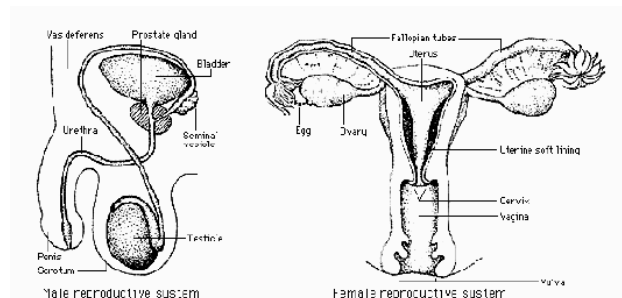
- An intricate network of nerves and often a central brain structure coordinate all systems and permit movement, thought, and interaction with the environment.

- The **NERVOUS SYSTEM** is the master controller of the body. Through its many parts, the nervous system monitors conditions both within and outside the body. It then processes that information and decides how the body should respond, if at all. This system sends out electrical signals that spur the body into immediate action. Although one of the smallest of the body's systems in terms of weight, the nervous system is the most complex and versatile.



➤ Reproductive System

- Reproduction is accomplished either sexually or asexually through specialized organs and specialized systems that ensure perpetuation of the species.



E. Reproduction, Life Cycles, and Heredity

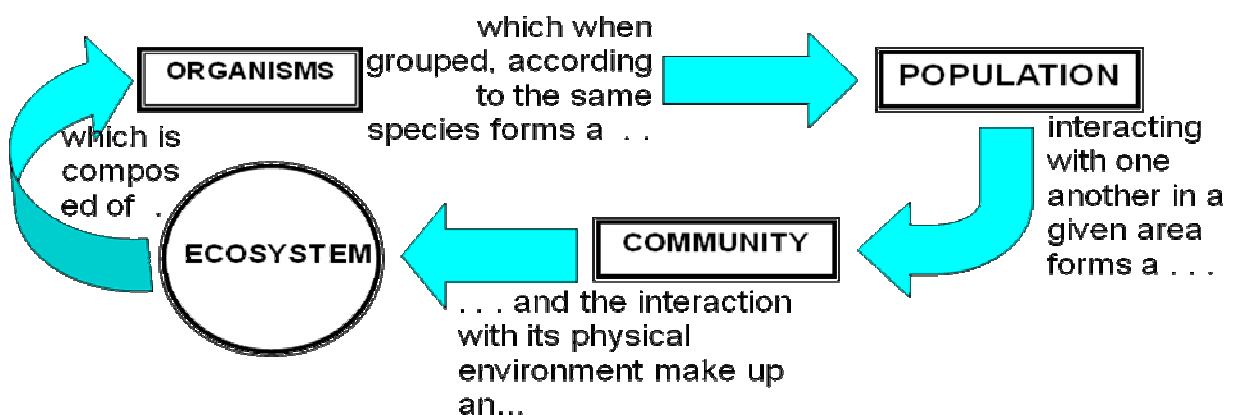
- Reproduction occurs in two ways: asexually and sexually.
 - In asexual reproduction, organisms are derived from only one parent (bacteria, algae, molds, fungi, many lower plants, protozoans, sponges, and coelenterates as well as in a few higher order animals).
 - Sexual reproduction requires the union of a male gamete (reproductive cell) and a female gamete.

- Chromosomes are made of genes with strands of DNA (deoxyribonucleic acid). DNA carries the code of life, which is the code for protein production.
 - Chromosomes come in pairs with a gene for each trait on each member of the pair; thus each trait is determined by two genes that may be dominant or recessive.
 - Dominant traits in humans include unattached earlobes, brown eyes, curly hair, polydactyly (more than 10 fingers or toes), and the ability to curl one's tongue.
 - Recessive traits include attached earlobes, blue or hazel eyes, straight hair, the normal number of fingers and toes, and the inability to curl one's tongue.
- Living organisms move through definite life cycles: they come into being, they grow, they may metamorphose, they mature, they reproduce, and they die.

Genetics and Evolution

- When gametes are formed, abnormal (or mutant) genes can be formed by erroneous duplication of the DNA, by physical transformation of the positions of molecules making up the DNA strands, or by external causes such as radiation, X-rays, or gamma rays.
- The basic principle of evolution is "survival of the fittest". This means that the individuals best adapted to the environment generally produce the greatest number of offspring.
- The theory of evolution offers an explanation for the immense diversity of life found on earth, through the process of natural selection; organisms adapt better to their environments.

G. Ecology



- Ecology is the study of the interactions of organisms with their environment and with each other. The environment in which living organisms exist is called the biosphere, and includes land, water, and air.
- The basic unit of ecological study is the ecosystem, a community of living organisms and their nonliving environment, interacting with each other. It is the environment through which energy flows and minerals recycle.
 - A balanced ecosystem has three fundamental characteristics: (1) there is a relatively constant source of energy (the sun); (2) The sun's energy is converted to glucose, which is needed by the living organisms; (3) Organic matter and nutrients are successfully recycled.

- Energy is transmitted through an ecosystem primarily by means of the food chain. Plants are the sources of all food in any ecosystem since they manufacture their own food. Some animals are herbivores, some are omnivores while others are carnivores. (energy follows a one-way path)
- Recycling of organic matter and nutrients within an ecosystem occurs as plants and animals excrete waste materials back into their environment and die. Microorganisms recycle all this material back to the environment. (biogeochemical cycles)
- Humans may upset the balance in ecosystems. Air and water pollution result to the death of some organisms. Increased levels of carbon dioxide may cause localized or global warming, upsetting the balance of energy. Increased CFC levels reduce the thickness of the ozone layer, causing increased levels of ultraviolet radiation.

WHAT FACTORS ARE INVOLVED IN THE INTERRELATIONSHIPS BETWEEN SCIENCE, TECHNOLOGY AND SOCIETY?

The scientifically literate person understands and appreciates the joint enterprises of science and technology, their interrelationships, and their impacts on society and the environment.

*** Science and Technology**

There is a distinction between science and technology, although they often overlap and depend on each other. Science deals with generating and ordering conceptual knowledge. Technology deals with design and development, and the application of scientific or technological knowledge, often in response to social and human needs. Technology does not just provide tools for science, however; it also may provide motivation and direction for theory and research. For example, the theory of the conservation of energy was developed technological problem of increasing the efficiency of commercial steam engines. The mapping of the locations of the entire set of genes in human DNA has been motivated by the technology of genetic engineering, which both makes such mapping possible and provides a reason for doing so. The invention of the microscope led to new discoveries about cells.

*** Scientists and Technologists are human**

Scientists see patterns in phenomena as making the world understandable; technologists see them as making the world manipulable. Scientists seek to show that theories fit the data; technologists (engineers) seek to demonstrate that designs work. Scientists cannot provide answers to all questions; engineers cannot design solutions for all problems. Outside of their specialized fields, scientists and technologists may not exhibit strong development of all or even most of the dimensions of scientific literacy. Vocations in science and technology are open to most people.

*** Impact of Science and Technology**

Scientific and technological developments have real and direct effects on every person's life. Some effects are desirable; others are not. Some of the desirable effects may have undesirable side effects. In essence, there seems to be a trade-off principle working in which gains are accompanied by losses. As society continues to

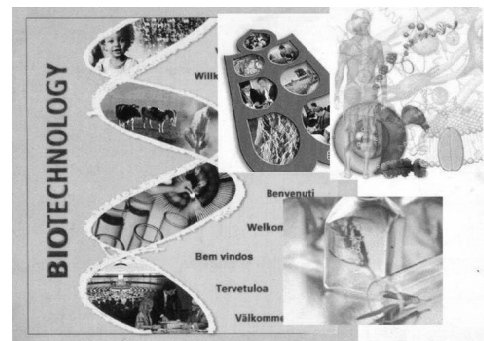


increase its demands on energy consumption and consumer goods, the people are likely to attain a higher standard of living while allowing further deterioration of the environment to occur. The different areas of technology represent areas of technological activity in wider world.

➤ **Materials Technology** – this involves the use and development of materials to achieve a desired result ... construction, preservation and recycling of materials.

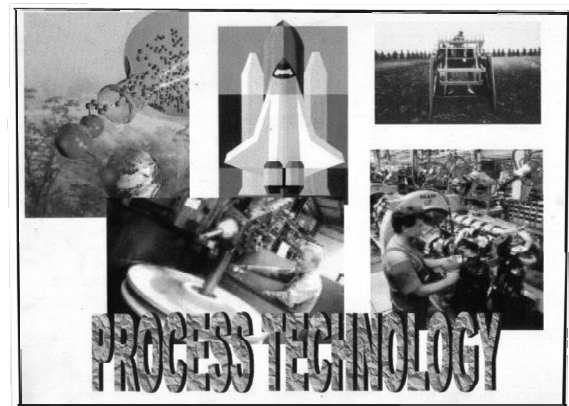
- The characteristics of natural materials could be changed by processing, such as the tanning of leather and the firing of clay. Plastics are synthesized and designed to make automobile and space vehicle parts, food packaging and fabrics, artificial hip joints and dissolving stitches.
- The disposal of used materials has become a problem. Some used materials, such as organic wastes, can be returned safely to the environment. But some materials, such as plastics, are not easily recycled nor do they decompose quickly when returned to the environment. Radioactive waste so hazardous for such a long time that how best to dispose of them is not clear.

➤ **Biotechnology** – involves agriculture, tissue culture, food production, health and the environment, industrial processes, genetic engineering



- The success of genetic engineering has helped to increase the natural variability within plant species by using radiation to induce mutations, so that there are more choices for selective breeding. For example, plants can be given the genetic program for synthesizing substances that give them resistance to insect predators.
- The widespread use of insecticides, herbicides, and fungicides has greatly increased useful farm output. However, their effectiveness may diminish as organisms develop genetically determined resistance to them, thereby requiring increased amounts of pesticides or the development of new ones.
- An alternative to the chemical control of pests is introducing organisms from other ecosystems in an effort to reduce the number of pests in the agricultural ecosystem (such as by using foreign insects that feed on local weeds).
- The development of rapid and cheap transportation reduces spoilage of food, as do treatment, additives, refrigeration, and packaging.

➤ **Process Technology** – this include chemical and industrial processes, mechanical systems, manufacturing, and transportation. Increasing automation requires less direct labor and fewer skilled crafts, but more engineering, computer programming, quality control, supervision, and maintenance. Automation reduces the workers' control and may eliminate some workers'



jobs even while it creates others. Flexibility and skill in learning a succession of new job roles have become increasingly important as the pace of technological change quickens.

- **Electronic/Control Technology** – this involves the design and production of simple circuits, robotics, and control systems, electronic devices.

- With the invention of devices to generate and control electric current, information could be encoded and conveyed over long distances by wires. With radio waves, the same information could be encoded as changes in wave pattern and distributed in all directions through the atmosphere without the need of connecting wires. The efficient control of light waves in lasers has made possible the encoding and transmitting of information as pulses in light intensity over optical fibers.
- Robots are used to perform the tasks of mass production. Instructions for processing are used to control the processes electronically, rather than having to be interpreted and carried out by people.



- **Information and Communication Technology**- this deals with locating and acquiring, storing, sorting, receiving communication information in variety of forms, data collection techniques, storage and data search programs, and generation models.

- Computers can provide control that is as good as, or more precise and rapid than, human control. The operation of automobile engines, the flight control of aircraft and spacecraft, and the aiming and firing of weapons can be computerized to take account of more information and to respond much more rapidly than a human operator could. However, there are also risks that the information entered may contain errors and its speed of response may exceed human ability to monitor or judge the output.
- Information storage also involves issues of privacy and security. Computer-managed information systems require means for ensuring that information cannot be changed or lost accidentally.

- **Health Technology** - is concerned with reducing the exposure of humans to conditions that threaten health, as well as with increasing the body's resistance to such conditions and minimizing the harmful effects that do occur.

- Sanitation measures include containment and disposal of garbage, construction of sewers and sewage processing plants, purification of water and milk supplies, quarantine of infectious patients, chemical reduction of insect and microorganism populations (insecticides and antiseptics), and suppression of the population of rats, flies, and mosquitoes that carry microorganisms.



- Vaccination has been the most effective means of preventing early death from disease. It is used to enhance the human body's natural defenses against disease. -- weakened or killed disease microorganisms injected into the blood may arouse the body's immune system to create antibodies that subsequently will incapacitate live microorganisms if they try to invade.
- The detection, diagnosis, and monitoring of disease are improved by different kinds of technology. A look inside the body has been provided by imaging devices that use slender probes to supply visible light or (from outside the body) magnetic fields, infrared radiation, sound waves, x rays, or nuclear radiation.
- Techniques for mapping the location of genes on chromosomes make it possible to detect disease-related genes in children or in prospective parents
- The modern treatment of many diseases also is improved by science-based technologies. Knowledge of chemistry, for example, has improved our understanding of how drugs and naturally occurring body chemicals work, how to synthesize them in large quantities, and how to supply the body with the proper dosage.
- Substances have been identified that are most damaging to certain kinds of cancer cells. Knowledge of the biological effects of finely controlled beams of light, ultrasound, x rays, and nuclear radiation (all at much greater intensities than are used for imaging) has led to technological alternatives to scalpels and cauterization.
- Tissue/ organ transplants make it possible to replace some body parts and to implant devices for electrically pacing the heart, sensing internal conditions, or slowly dispensing drugs at optimal times.
- Improved medical technologies raise ethical and economic issues. The combined results of improved technology in public health, medicine, and agriculture have increased human longevity and population size thus increasing the challenge of providing all humans with adequate food, shelter, health care, and employment, and it places ever more strain on the environment.
- The developing technology of diagnosing, monitoring, and treating diseases and malfunctions increases society's ability to keep people living when they otherwise would have been unable to sustain their lives themselves.
- There is continuing debate over abortion, intensive care for infants with severe disabilities, maintaining the life functions of people whose brains have died, the sale of organs, altering human genes, and many other social and cultural issues that arise from biomedical technology.



PRACTICE TEST

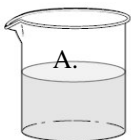
Direction: Analyze the following questions and choose the letter of the best answer:

1. Which of the following is an orderly, systematic approach to problem solving?
A. conclusion
B. hypothesis
C. experiment
D. scientific method
2. When two equal forces act on the same object in opposite directions, the net force is
A. smaller than either force.
B. equal to each of the forces.
C. zero.
D. greater than either force.
3. The greater the mass of an object,
A. the easier the object starts moving.
B. the greater its inertia.
C. the more balanced it is.
D. the more space it takes up.
4. According to Newton's third law of motion, when a hammer strikes and exerts force on a nail, the nail
A. exerts an equal force back on the hammer.
B. creates a friction with the hammer.
C. moves at a constant speed.
D. disappears into the wood.
5. Water boils, therefore if a liquid boils it must be water. This statement is :
A. correct, because boiling water is a liquid.
B. correct, because water is a liquid that boils.
C. wrong, because water does not always boil.
D. wrong, because not all liquids is water.
6. What is the correct order for steps in the scientific method?
A. gather data, formulate hypothesis, have a question, develop method, draw conclusion
B. formulate hypothesis, have a question, draw conclusion, gather data, develop method
C. have a question, formulate hypothesis, develop method, gather data, draw conclusion
D. develop method, have a question, formulate hypothesis, draw conclusion, gather data
7. Which of these substances is the LEAST dense?
A. wood
B. copper
C. mercury
D. rubber
8. For work to be done on an object,
A. the object must not move.
B. some force need only be exerted on the object.

- C. the object must move some distance as a result of a force.
 - D. the object must move, whether or not a force is exerted on it.
9. In order to do work on an object, the force you exert must be
- A. in a direction opposite to Earth's gravitational force.
 - B. the maximum amount of force you are able to exert.
 - C. in the same direction as the object's motion.
 - D. quick and deliberate.
10. Which of the following statements is NOT TRUE of energy in the biosphere?
- A. Energy flows a one-way path
 - B. Energy is conserved in the biosphere
 - C. Energy follows a cycle through the web of life
 - D. Energy leaves the living system in the form of heat
11. Without friction there would be
- A. less machine efficiency.
 - B. greater output work than input work.
 - C. greater input work than output work.
 - D. equal input and output work.
12. An example of something that stores chemical energy is
- A. lightning.
 - B. a microwave.
 - C. a match.
 - D. light.
13. What type of conversion is taking place when natural gas is used to heat water?
- A. chemical energy into thermal energy
 - B. thermal energy into mechanical energy
 - C. mechanical energy into electromagnetic energy
 - D. electromagnetic energy into chemical energy
14. When you rub your hands together on a cold day, you use friction to convert
- A. thermal energy into nuclear energy.
 - B. nuclear energy into electrical energy.
 - C. mechanical energy into thermal energy.
 - D. electrical energy into electromagnetic energy.
15. The type of energy stored by fossil fuels such as coal is
- A. kinetic energy.
 - B. mechanical energy.
 - C. chemical potential energy.
 - D. electromagnetic energy.
16. Which statement is TRUE of gases?
- A. The particles that make up gases are packed together in a relatively fixed position.
 - B. Gases expand to fill all the space available.
 - C. Gases have a definite volume.
 - D. Gases have a definite shape.
17. Particles of a liquid
- A. have no viscosity.
 - B. decrease in volume with increasing temperature.
 - C. are tightly packed together and stay in a fixed position.

- D. are free to move within a container but remain in close contact with one another.
18. Water vapor in the air turns to liquid water in the form of rain. This is an example of a
- A. physical change.
 - B. chemical change.
 - C. chemical equation.
 - D. chemical formula.
19. Which is a characteristic property of acids?
- A. Acids turn blue litmus paper red.
 - B. Acids turn red litmus paper blue.
 - C. Acids taste bitter.
 - D. Acids do not react with metals.
20. Which of the following statements is NOT part of the cell theory?
- A. Cells are the basic unit of structure and function in living things.
 - B. All cells are produced from other cells.
 - C. All living things are composed of cells.
 - D. Only animals are composed of cells.
21. What is the function of a cell wall?
- A. to protect and support the cell
 - B. to prevent water from passing through it
 - C. to perform different functions in each cell
 - D. to prevent oxygen from entering the cell
22. The function of a cell membrane is to _____.
- A. to protect and support the cell
 - B. to control what enters and leaves the cell
 - C. to perform different functions in each cell
 - D. to form a hard outer covering for the cell
23. What happens during photosynthesis?
- A. The cell uses oxygen to make food.
 - B. The cell uses glucose to make oxygen.
 - C. The cell uses the energy in sunlight to make food.
 - D. The cell uses the energy in sunlight to make carbon dioxide.
24. What happens during respiration?
- A. Oxygen is released into the air.
 - B. Glucose is broken down, releasing energy.
 - C. Carbohydrates are released into the bloodstream.
 - D. Water and carbon dioxide are converted into energy.
25. A change to the environment that has a negative effect on living things is called
- A. a renewable resource.
 - B. a nonrenewable resource.
 - C. population growth.
 - D. pollution.
26. In an energy pyramid, which level has the most available energy?
- A. producer level
 - B. first-level consumer level
 - C. second-level consumer level
 - D. third-level consumer level

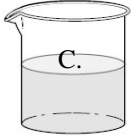
27. When a flea is living on a dog, the dog is the
A. parasite. C. predator.
B. host. D. prey.
28. The *Escherichia coli* that live in your intestine and help break down food are an example of which type of interaction?
A. competition. C. parasitism.
B. mutualism. D. predation.
29. Suppose that you close a window because you notice that rain is falling. Your action in closing the window is a(n)
A. involuntary reaction. C. stimulus.
B. depressant. D. response.
30. Coal is NOT classified as a mineral because
A. it is useful as a fuel.
B. it comes from the remains of plants and animals.
C. it is not valuable.
D. it is part of Earth's crust.
31. What is the correct order (starting from the surface) of Earth's layers?
A. crust, outer core, inner core, mantle
B. mantle, outer core, inner core, crust
C. crust, mantle, outer core, inner core
D. outer core, inner core, crust, mantle
32. Which of the following environmental conditions has been blamed for the usual occurrence of "fish kills"?
A. Acid rain C. Atmospheric pollution
B. Eutrophication D. Silting of lakes
33. Fossil fuels are considered nonrenewable resources because they
A. pollute the air.
B. burn so quickly.
C. are in such high demand.
D. take hundreds of millions of years to form.
34. What is the most widely used source of renewable energy in the world today?
A. hydroelectric power C. biomass fuels
B. solar power D. nuclear power
35. How is the conservation of mechanical energy described in a falling object?
A. A falling object loses kinetic energy and gains potential energy.
B. A falling object loses kinetic energy and loses potential energy.
C. A falling object gains kinetic energy and loses potential energy.
D. A falling object gains kinetic energy and gains potential energy.
36. Which law of motion explains what happens every time we walk?

- A. Universal gravitation
B. Interaction
C. Acceleration
D. Inertia
37. What is the escape mechanism by which the mud fish and the catfish change from a darker color to a lighter one depending on the color of the water called?
A. protective mimicry
B. adaptive coloration
C. habitat protection
D. aggressive coloration
38. If the earth will stop spinning on its axis, which will likely to happen?
A. There will be an increase in the number of days in a year.
B. The clouds will start to block the sun's shadow.
C. The sun will start to revolve around the earth.
D. There will be no night and day.
39. Which sentence best describes a scientific theory?
A. It can never change.
B. It can be proven conclusively.
C. It can change every time it is tested.
D. It is well-tested and explains a wide range of observations.
40. Which of the following phrases best describes science?
A. doing experiments
B. drawing conclusions
C. a list of facts about the world around you
D. a process of learning about the natural world
41. Although science and technology complement each other, they are different from each other. Which of two statements below clearly show the difference between science and technology?
1. Science is a way of knowing; technology is a way of doing.
2. Science is a method of answering theoretical questions; technology is a method of solving practical problems.
3. Science has to do with techniques and procedures for finding solutions to problems; technology has to do with discovering relationships among variables.
4. Advances in science are measured in terms of human factor; in technology these are measured in terms of creative design and originality.
A. 1 & 2
B. 2 & 3
C. 3 & 4
D. 1 & 4
42. In which beaker will sugar dissolve easily?
- 

A.
Water at 0°C



B.
Water at 15°C



C.
Water at 30°C

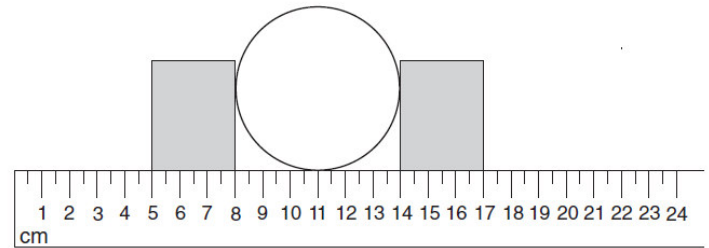


D.
Water at 60°C
- A. A
B. B
C. C
D. D
43. The nonliving parts of an ecosystem are called
A. populations.
C. biotic factors.

B. organisms.

D. abiotic factors.

44. The diagram at the right shows a ball in place by two blocks. The diameter of the ball is shown between the blocks. What is the diameter of the ball?



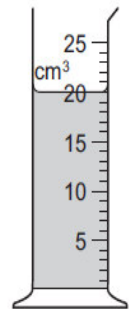
- A. 6.0 cm
- B. 8.0 cm
- C. 14.0 cm
- D. 17.0 cm

45. Which energy changes takes place in a flashlight with batteries?

- | | | | | |
|---------------|---|------------|---|------------|
| A. chemical | → | electrical | → | light |
| B. electrical | → | light | → | chemical |
| C. light | → | chemical | → | electrical |
| D. chemical | → | light | → | electrical |

46. The diagram at the right shows a liquid in a graduated cylinder. The mass of the liquid is 16 g. What is the density of the liquid?

- | | |
|--------------------------|---------------------------|
| A. 320 g/cm ³ | C. 1.25 g/cm ³ |
| B. 36 g/cm ³ | D. 0.8 g/cm ³ |



47. Which skill are you using if you identify a trend or pattern seen in a graph?

- | | |
|----------------------|--------------------------|
| A. inferring | C. posing questions |
| B. interpreting data | D. developing hypotheses |

48. Which of these is an example of work being done on an object?

- A. holding a door shut on a windy day so it doesn't blow open
- B. trying to push a car that doesn't move out of a road hole
- C. holding a heavy piece of wood at a construction site
- D. pushing a child on a swing

49. Which of the following phrases best describes science?

- A. doing experiments
- B. drawing conclusions
- C. a list of facts about the world around you
- D. a process of learning about the natural world

50. Which is the correct set of base properties?

- A. Sour taste, corrosive, change litmus from red to blue
- B. Bitter taste, corrosive, change litmus from blue to red
- C. Bitter taste, slippery, change litmus from red to blue
- D. Sweet taste, slippery, change litmus from blue to red